13 Consolidation, starter routines and transition to Year 12 (Première)

Sheet 75 Tue 1st June 2027 *Synthesis problems drawing on several topics*

1. Warm-up · Quick [EQ] Solve $3(x-1) = x+5$.
   Ans. $3x-3 = x+5$, $2x = 8$, $x = 4$.
2. Core · Method [GEO] For which value of k are the points $A(1,1)$, $B(3,5)$ and $C(k,9)$ collinear?
   Ans. $\overrightarrow{AB}\begin{pmatrix}2\\4\end{pmatrix}$, $\overrightarrow{AC}\begin{pmatrix}k-1\\8\end{pmatrix}$: $2\times 8 - 4(k-1) = 0$, so $k = 5$.
3. Core + · Method [PRO] A price increases by 10% then decreases by 10%. What is the overall change?
   Ans. $1.1 \times 0.9 = 0.99$: a decrease of 1%.
4. Review · Quick [PER] Total surface area of a cuboid with dimensions 4, 3, 2 (cm)?
   Ans. $2(4\times 3 + 4\times 2 + 3\times 2) = 52$ cm^2.
5. Thread · Quick [SOL] Calculate the exact volume of a sphere of radius 3 cm.
   Ans. $V = \dfrac{4}{3}\pi \times 27 = 36\pi$ cm^3.

Sheet 76 Thu 3 June 2027 *Review of the six competences*

1. Warm-up · Metacognition [MET] List the six mathematical competences developed at lycée.
   Ans. Investigate, model, represent, reason, calculate, communicate.
2. Core · Metacognition [MET] Match each of the following to a competence: "translating a problem statement into an equation"; "sketching a curve"; "justifying an equality with a written calculation".
   Ans. Model; represent; reason (or communicate, depending on the emphasis placed on the write-up).
3. Core + · Metacognition [MET] To solve $2x + 3 = 11$, which competence is mainly used?
   Ans. Calculate (manipulating a linear equation).
   For oral discussion: setting up an equation from a problem comes under "model".
4. Review · Problem [MOD] Water at 20 °C is heated by 3 °C per minute: $T(t) = 20 + 3t$. After how many minutes does it reach 65 °C?
   Ans. $20 + 3t = 65 \Leftrightarrow t = 15$ min.
5. Thread · Quick [EQ] Solve in $\mathbb{R}$: $0.5x + 1.2 = 2.7$.
   Ans. $x = 3$

Sheet 77 Mon 7 June 2027 *Targeted reactivation of prerequisites for Première*

1. Warm-up · Quick [DEV] Expand $(x+3)^2$.
   Ans. $x^2 + 6x + 9$.
2. Core · Method [EQ] Check that $x^2 - 5x + 6 = (x-2)(x-3)$ and deduce the solutions of $x^2 - 5x + 6 = 0$.
   Ans. $(x-2)(x-3) = x^2 - 5x + 6$; zero product: $x = 2$ or $x = 3$.
3. Core + · Anticipation [MOD] For $f(x) = x^2$, calculate the rate of change $\dfrac{f(3) - f(1)}{3 - 1}$.
   Ans. $\dfrac{9-1}{2} = 4$: this prepares the idea of the derivative in Première.
   Première topic, simple consolidation calculation.
4. Review · Construction [CON] Place M and N, the midpoints of $[AB]$ and $[AC]$, then draw $[MN]$. What can be said about (MN) and (BC)? Calculate MN if $BC = 9$ cm.
   Ans. $(MN) \parallel (BC)$ and $MN = \dfrac{9}{2} = 4.5$ cm.

5. Thread · Write-up [DEM] Prove that a triangle with sides 7, 24 and 25 is right-angled.

Ans. $7^2 + 24^2 = 49 + 576 = 625 = 25^2$: converse of Pythagoras' theorem.

Sheet 78 Tue 8 June 2027 *Synthesis problems: choosing and justifying a strategy*

1. Warm-up · Reading code [ALG] What does $f(2)$ return?

```
def f(x):
    return x**2 - 3*x + 1
```

Ans. $4 - 6 + 1 = -1$.

2. Core · Completion [ALG] Complete the code so that the function returns `True` when A, B, C are collinear:

```
def alignes(xa, ya, xb, yb, xc, yc):
    return (xb-xa)*(yc-ya) - (yb-ya)*(xc-xa) == ....
```

Ans. We compare the determinant with 0: `== 0`.

3. Core + · Completion [ALG] Complete the absolute value function:

```
def vabs(x):
    if x >= 0:
        return x
    else:
        return ....
```

Ans. `return -x`.

Check: `vabs(-4)` returns 4.

4. Review · Quick [NUM] Write 2.5×10^3 and then 6.2×10^{-2} in decimal notation.

Ans. 2,500; 0.062.

5. Thread · Quick [REP] Give the reflection of $A(2, -3)$ in the origin.

Ans. $(-2, 3)$.

Sheet 79 Thu 10 June 2027 *Presentation of the specialist subjects; review*

1. Warm-up · Quick [NUM] Express $\dfrac{3}{8}$ as a percentage.

Ans. 37.5%.

2. Core · Metacognition [MET] What are your three revision priorities before Première?

Ans. Personal answer: identify, in this year's sheets, the questions where you hesitated the most.

3. Core + · Quick [EQ] Solve $2x^2 = 18$.

Ans. $x^2 = 9$: $x = 3$ or $x = -3$.

4. Review · Trap [PUI] Calculate $(-2)^4$ and -2^4.

Ans. 16 and -16.

The brackets decide what the exponent applies to.

5. Thread · True/false [DIV] True or false? Justify. If an integer is divisible by 4 and by 6, then it is divisible by 24.

Ans. False. Counter-example: 12 is divisible by 4 and by 6, but not by 24; it is divisible by $\operatorname{lcm}(4, 6) = 12$.